

HEALTHCONNECT CLINIC

Week 7 Project Summary | Data Analytics Track

AnalystLab Africa Experience Lab Internship Programme

Project	HealthConnect Clinic Experience Lab
Project Title	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Week	Week 7
Track	Data Analytics
Status	Testing, Refinement & Validation

Project Overview

Week 7 continued directly from the Week 6 Advanced Analytics and Decision Support work. The focus moved from development and integration into systematic testing, refinement, re-testing and validation. The existing HealthConnect appointment dataset, Power BI model and Week 6 findings were retained.

Testing Focus

The work focused on validating KPI calculations, checking dashboard values against the underlying data, testing filters and calculations, investigating inconsistent behaviour, refining weaknesses, re-testing revised outputs, reassessing key findings across relevant segments, and preparing the Analytics contribution for Week 8.

Area	Work Completed	Outcome
Core KPIs	Overall No-Show Rate; 30+ Day No-Show Rate; Repeat No-Show Rate; Lost Slot Rate	Validated
Lead time	0–3, 4–14, 15–30 and 30+ day segments	Validated
Previous no-show history	Has prior no-show / No prior no-show filters	Validated after refinement
Reminder channel	Email, None, SMS and WhatsApp	Validated
Appointment type	Four appointment-type segments	Validated
Dashboard functionality	Slicers, KPI labels, layout and filter behaviour	Refined and re-tested
Distance to clinic	Category sort order in the Lead Time × Distance chart	Refined and re-tested

Most Important Testing Outcome

The main substantive issue identified was a filter-context problem in the Repeat No-Show Rate measure. When the Previous No-Show History slicer was set to “No prior no-show”, the original measure continued to return 55.4% because its DAX explicitly filtered on the “Has prior no-show” category rather than responding to the slicer selection. The measure was refined to respect the selected history context. Re-testing returned 0.0% for the “No prior no-show” selection, confirming the correction.

Dashboard Refinements

- Rearranged the Week 6 dashboard to improve analytical flow and make space for interactive analysis.
- Removed the static Top 3 Insights panel to create space for slicers; the underlying findings were retained in the documentation.
- Added slicers for Lead Time, Previous No-Show History, Reminder Channel and Appointment Type.
- Renamed the second KPI to “30+ Day No-Show Rate” to make its segment-specific meaning clear.
- Standardised user-facing labels and lead-time categories.

- Corrected the distance-to-clinic category sort order to a logical distance progression (<5km, 5-15km, 15km+, Unknown) using a custom sort column in Power Query.
- Retained the analytical findings while improving dashboard usability.

Validated Key Findings

Finding	Validated Result	Interpretation
Overall no-show rate	48.5%	Validated baseline across 5,000 appointments.
Booking lead time	24.8% → 60.5%	Observed no-show rate increased across the tested lead-time bands.
30+ days + prior no-show	67.9%	Highest observed combined segment; association, not causation.
30+ days + no prior no-show	55.2%	Long lead time remained elevated without previous no-show history.
Lost Slot Rate	53.7%	No-show and cancelled appointments together represented substantial scheduled-capacity loss.

Cross-Track Collaboration

The Week 6 Data Analytics–Data Science collaboration was carried forward into Week 7. Analytics revalidated the Lead Time × Previous No-Show finding through dashboard and filter testing. The previously exchanged finding was that the 30+ day group with prior no-show history had a 67.9% observed no-show rate, compared with 55.2% without prior no-show history. Data Science had identified an explicit interaction feature as a candidate for model testing.

No further Data Science model-performance response was available at the time of submission. The interaction is therefore documented as a pending cross-track dependency, not as a completed model improvement.

Challenges, Decisions and Remaining Issues

Issue / Observation	Action / Decision	Status
Repeat No-Show KPI ignored the “No prior no-show” slicer	Refined DAX and re-tested	Resolved
30+ KPI remained 60.5% under other lead-time selections	Retained because it is a segment-specific KPI	Expected behaviour
0–3 Lost Slot Rate re-checked against raw data	Confirmed exact match (31.45% rounds to 31.4%); no discrepancy found	No issue — confirmed correct
Distance-to-clinic category order did not follow a logical progression	Applied custom sort order in Power Query; re-tested	Resolved
Further Data Science response was pending	Documented as an open dependency; no model-improvement claim made	Open

Contribution to the Overall HealthConnect Project

The Week 7 Data Analytics contribution strengthened confidence in the existing analytical component by validating KPI calculations, testing interactive dashboard behaviour, resolving a genuine filter-context issue, correcting a chart presentation issue, improving dashboard usability, and carrying a validated high-risk segment forward as a clearly defined cross-track modelling dependency.

Limitations

- The dataset is fictional and synthetic and is intended for learning and analytical practice rather than direct clinical decision-making.
- The analysis identifies observed associations and does not establish causation.
- Distance and waiting-time fields contain missing values; waiting time is treated cautiously because values are present for many no-show and cancelled appointments.
- Reminder-channel differences should not be interpreted as proof that one channel causes better attendance.
- The Data Science interaction feature remains a candidate for predictive testing.

Week 8 Readiness

- Use the refined Power BI dashboard and validated KPI logic, including the corrected distance sort order, as the current Analytics version.
- Carry the validated findings and documented limitations into final integration.
- Complete or document the status of the Data Science interaction-feature test before final presentation, where applicable.

Evidence Reference

Detailed screenshots and test records, including evidence of the distance-to-clinic sort-order fix, are maintained in the separate Week 7 Testing & Validation Evidence Report. The cross-track activity is documented separately in the Week 7 Cross-Track Testing & Validation Evidence document.